



DURBAN UNIVERSITY OF TECHNOLOGY
INYUVESI YASETHEKWINI YEZOBUCHWEPHESHE

FACULTY OF ACCOUNTING & INFORMATICS

**DEPARTMENT OF INFORMATION
TECHNOLOGY**

PROGRAMME TITLE

***Bachelor of Information and Communications Technology
&
Advanced Diploma in Information and Communications
Technology***

PROGRAMME CODE

BINCT1 & ADICT1

Study Guide

2026

MODULE / SUBJECT TITLE

Platform Based Development

MODULE / SUBJECT CODE: PBDV301 | PBDE401

HEQSF LEVEL: 7

HEQSF CREDITS: 16

FACULTY OF ACCOUNTING & INFORMATICS.....	1
General Information.....	3
1. Welcome.....	4
1.1 Faculty and Departmental Policies	4
1.2 Subject/Module Policies	4
2. Using your onlineThinkLearnZone (Moodle) classroom.....	7
3. Introduction to the Module	8
4. Learning Outcomes of Platform Based Development.....	8
5. Learning, teaching and assessment strategies	9
6. Learning and teaching methods:	9
5.2 Learning and teaching strategies	9
5.3 Learning activities	10
5.4 Graduate attributes	10
5.5 Library Orientation	10
6. Scheme of Work.....	11
7. Copyright and Plagiarism.....	12
8. Student Support	12
9. Technology Advisory Board (ITAB).....	13
10. Quality Assurance and Enhancement.....	13

General Information

Module Co-ordinator	:	Mr Freedom Khubisa
Lecturers	:	Name : Freedom Khubisa Telephone : 031-373-5543/96 e-Mail : freedomk@dut.ac.za
Offices	:	Ritson Road Campus Department of Information Technology 2nd Floor, West Wing,
Consultations Times	:	To be announced during lecture time.
HoD Office	:	Mr. Ebrahim Asmal Ritson Road Campus, West Wing Department of Information Technology: 2nd Floor
Telephone	:	031-373-5565/94
Secretaries	:	Name : Ms Thabisile Ntuli Telephone : 031 373 5594 e-Mail : thabilenl@dut.ac.za
Admin Assistant	:	Name : Ms Abongile Cele Telephone : 031 373 5596 e-Mail : itdept@dut.ac.za / abongileC@dut.ac.za
Lectures	:	3 per week: theory lectures (hour each)
Lecture Venue	:	As indicated on the official timetable.
Tutorials Moodle)	:	1 per week: one hour practical (Also check TLZ Moodle)
Course Duration	:	Approximately 14 weeks

Relevant Policies and rules: Please refer to the Departmental and Faculty handbooks for all institutional policies, rules and procedures. Students are encouraged to familiarise themselves with these policies and to adhere to them strictly.

I. Welcome

Congratulations on successfully registering for the Bachelor of Info & Com Technology or Advanced Diploma in ICT and a hearty welcome to Platform Based Development module. The primary objective of this module is to equip students with the ability to construct a clear understanding of how to develop software's and applications for various platforms.

The purpose of this learner study guide is to provide students with a comprehensive overview of the Platform Based Development subject matter. This will enable students to anticipate the module's content and how the learning outcomes will be assessed.

In addition, this guide outlines the essential learning outcomes that students must achieve and demonstrate by the end of this module. It also includes information on the learning and teaching strategies that will be employed, as well as the assessment and grading process. A list of recommended readings is provided, along with a scheme of work to assist students in planning their studies for the module.

Finally, specific rules and policies are outlined to ensure that students have a clear understanding of the module's expectations and requirements.

I.1 Faculty and Departmental Policies

Please refer to the Departmental and Faculty handbooks for all institutional policies, rules and procedures. Students are encouraged to familiarise themselves with these policies and to adhere to them strictly.

I.2 Subject/Module Policies

Academic Integrity

All work submitted must be your own. Collaboration is encouraged, but plagiarism or unauthorized assistance will result in **disciplinary action**. Use of open-source code/ work must be properly cited.

Participation and Attendance

Active participation in all module activities, including lectures, labs, and group projects, is expected. Attendance is **mandatory** for all classes since lectures usually cover, at least in outline, the essential contents of each section and give guidance with regards to further readings, etc. Furthermore, announcements – for example, about test dates or any imperative changes, are usually made during the lecture from time to time.

The lecturer also provides students with an opportunity to share their views and engage with other classmates to discuss interesting problems. The essence of a good lecture is good participation. A co-operative approach (from you and the lecturer) is encouraged. If you miss lectures, your lecturer will expect you to make up the work by yourself. It is, thus, very important for students to be on time and attend lectures and tutorials to be successful. Please note that any absences must be justified.

Any clash with other lectures has to be resolved by the student, it is **NOT** the lecturer's responsibility to cover material that a student has missed because of clashes (clashes will only occur if a student has not passed a course from the previous level).

Punctuality and Inappropriate Behaviour

Students are expected to be punctual i.e. arrive on time for lectures. Late coming will **NOT** be tolerated, because this is a disturbance. Inappropriate behaviour may include arriving late, leaving early, continuously talking, or any of the above. Rude and inappropriate behaviour will **NOT** be tolerated.

Student/Lecturer Communication Protocol

Students should use lecture time for questions as much as feasible. Students should also take use of consultation hours. The lecturer will announce module consultation times (if not provided on this guide). Students can schedule further consultations provided that they have attended ALL lectures and completed all work necessary.

Students should follow the escalation protocol to communicate valid concerns during their studies in a structured and efficient manner:

1. First Contact—Class Representative

If you have a reasonable issue about the course, its content, delivery, or anything else, please inform your class representative. The class representative is the main interface between students and the lecturer, ensuring your issues are heard.

2. Escalation — Lecturer

After receiving your complaint, the class representative will notify the lecturer. This stage allows the lecturer to immediately address and fix the issue, assuring rapid and effective resolution.

3. Escalation—Year and Programme Coordinators

If the lecturer does not resolve your issue, the class representative will escalate to the subsequently the Programme Coordinator (PC). This allows for more monitoring and possible solutions by reviewing the issue at a higher level.

4. Escalation—Department Head (HoD)

The class representative or Lecturer or PC will escalate the problem to the HoD PC fail to resolve it. The HoD reviews and addresses all concerns in accordance with department policies and procedures.

Assessment and Feedback

Assessments will be conducted through a mix of quizzes, spot tests, formal tests, and project work. The dates for assessments are indicated in Table I. Announcements regarding any adjustments to these assessments will be disseminated during lectures or via Moodle TLZ.

Table 1: Assessment Information

Assessments	Assessment Engagement Mode	Due-Dates
Web-based application project	Group Assignment	
Test 1	Individual	19-26 March 2026
Mobile-based application project	Group Assignment	
Test 2	Individual	28 April – 6 May
IoT / Game application project	Group	
Industrial project	Individual	

Feedback will be provided at least within 10 days of submission. This will be in the form of assessment mark as well as the learner answer script/marking rubric. Students are encouraged to seek additional feedback during consultation hours.

Any lecture can include “quizzes and spot tests” in addition to scheduled assessments. The only valid justification for missing any assessment is a doctor's, hospital, funeral, or prison note. The lecturer must receive this document within **three days** of the assessment date. The lecturer may call the document's authorizing organization to verify its details. Student must inquire if there will be granted an opportunity to write the missed assessment and gather any other pertinent details.

Technology Use

Students must adhere to the university's acceptable use policy for all module-related digital platforms and resources. The following are platforms that will be utilized in our teaching and learning:

1. **TLZ Moodle platform** – Course material such as recommended textbooks for various module content, research articles, videos, study guides, project rubrics, and so forth, will be shared via this platform.
2. **MS Teams** – This platform may be utilized for engagements, consultations, and facilitating presentations, in addition to any other suitable course component.
3. **Emails** – Formal correspondence will exclusively be conducted via DUT emails (ONLY DUT email addresses will be given attention).

You will therefore need to ensure that you are enrolled in Platform Based Development class on Moodle TLZ to be able to access all relevant module content.

Assessment Strategies

The evaluation of learning outcomes will occur through seven (7) mandatory assessments during the semester; an end-of-semester examination will NOT be administered. The final mark for this module is composed of:

Assessment	Percentage (%)
Project 1 – for web application	
Project 2 – for mobile application	
Project 3 – for game application	
Quizzes & Spot Tests	
Test 1	
Test 2	
Tutorials	

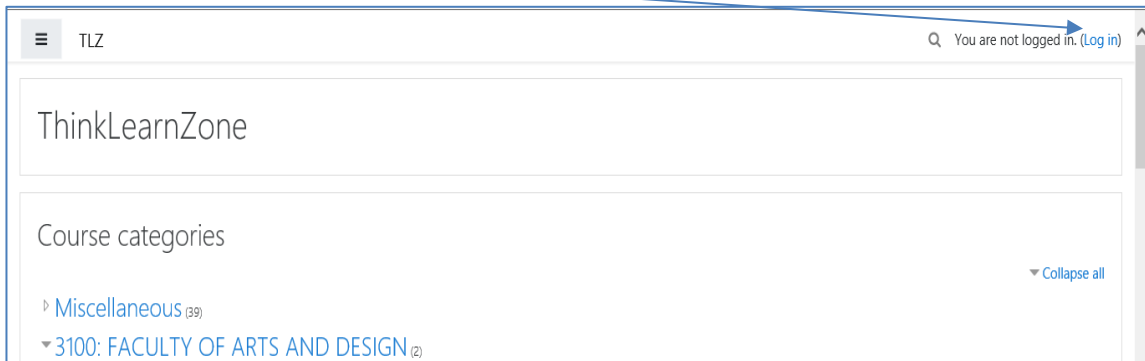
Other

Lecturers may deem it fit to introduce some other internal class/groups policies to allow for the smooth running of lectures.

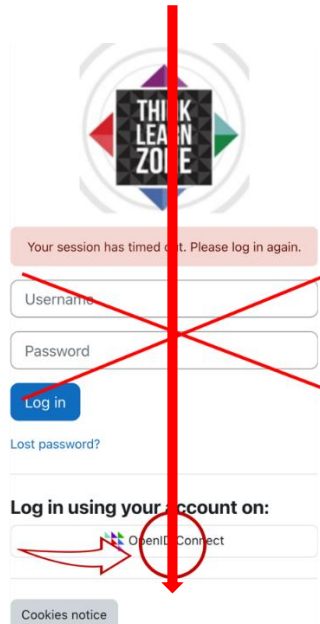
2. Using your online ThinkLearnZone (Moodle) classroom

All taught subjects/modules have their own online classroom on the ThinkLearnZone. You can access your classroom at <https://tlzprod.dut.ac.za>

Click on (Log in) on the top right corner



On the next screen click on “OpenID Connect”



Use your dut4life email address to sign (yourstudentnumber@dut4life.ac.za) and your email password as your password (Dut + first six digits of your ID no., e.g. Dut980610. For first-time 2020 students - use \$\$Dut + first six digits of your ID no., e.g. \$\$Dut980610)

Ask your lecturer for guidance or check out “how to log in” from here: <https://www.dut.ac.za/faq-students/>

The e-learning website also has contact information for help and technical assistance <https://www.dut.ac.za/elearning-contacts/>

3. Introduction to the Module

Welcome to the Platform Based Development module (PBDV301 | PBDE401) - we're glad to have you on board! The primary objective of this module is to provide students with the skills to develop software and applications for various platforms. Developing a new software application involves many decisions, with the first decision relating to the platform on which the application will run. There are numerous platforms available, including web-based, games, industrial, and mobile platforms, with popular mobile platforms including Android, iOS, and Windows. Each platform comes with its own benefits and challenges, and understanding the constraints and opportunities of available platforms is the focus of this module.



Once the platform decision is made, another critical decision is required: what technologies should be used to develop the proposed software or application? The choice of platform usually dictates the range of technologies that can be used for development. This module will introduce students to the development technologies used by major software companies and commonly adopted worldwide. Examples of such technologies include HTML5, CSS3, JavaScript, Typescript, Android Studio, Android SDK, Apple XCode I I, Flutter, Unity, Unreal Engine, GODOT, JQuery Mobile, and more. It is essential for students to understand that technology decisions must take into account both the functional and non-functional requirements of a given software or application.

4. Learning Outcomes of Platform Based Development

1. Demonstrate an understanding of platforms and associated languages
2. Compare industrial platforms and architecture
3. Develop a simple web application
4. Develop a simple mobile application for a given mobile platform
5. Develop a simple application on a game platform

5. Learning, teaching and assessment strategies

6. Learning and teaching methods:

This module adopts a diverse range of learning and teaching methods designed to cater to the varied learning preferences of our students and to equip them with the skills and knowledge required for success both in academia and in their future careers. Below is an overview of the teaching methods you could experience:

1. **Lecture with Discussion:** Traditional lectures are enriched with interactive discussions, encouraging students to engage with the material and participate actively in their learning process.
2. **Cooperative Learning:** This approach emphasizes teamwork and collaboration among students, allowing them to learn from each other and tackle complex problems collectively.
3. **Practical Work:** Hands-on sessions in labs provide students with the opportunity to apply theoretical knowledge to real-world scenarios, enhancing their practical skills.
4. **Individual Work:** Students are tasked with individual work and assessments, promoting independence and self-driven learning, critical for personal development and problem-solving skills.
5. **Group Work:** By working in groups, students develop their teamwork and communication skills, learning to collaborate effectively and share ideas.
6. **Guest Lecturing by Industry Experts:** Inviting professionals from various industries enriches the curriculum with current trends and real-world insights, bridging the gap between academic learning and industry practices.
7. **Presentations:** Through presentations, students learn to effectively communicate their ideas, refine their public speaking skills, and receive constructive feedback from peers and lecturers.

5.2 Learning and teaching strategies

By incorporating a variety of resources and materials, we aim to cater to diverse learning styles and preferences, ensuring that each student has access to the tools they need to succeed. The strategies include:

- **Study Guides** - Tailored study guides provide structured pathways through course material, offering direction and highlighting key concepts to focus on, facilitating effective learning.
- **Recommended Textbooks**- A carefully selected list of textbooks forms the core of our curriculum, supplemented by additional recommended readings to deepen understanding and broaden knowledge.
- **Videos**- Educational videos are utilized to present information in a dynamic and engaging format, making complex subjects accessible and stimulating interest.
- **Computer Journals**-Access to the latest computer journals and magazines keeps students abreast of technological advancements and trends in the field, fostering a culture of continuous learning.

5.3 Learning activities

For this module, students are allocated a substantial commitment, reflected in a credit value of 16 and encompassing 160 notional hours. These hours are a comprehensive measure, designed to encapsulate the entirety of the learning experience — from attending lectures and participating in discussions to engaging in individual study and completing assessments. This allocation ensures a thorough immersion in the module content, facilitating deep understanding and mastery of the subject matter.

Learning activity	% learning time
Contact (lectures)	40
Practical	20
Tutorial	20
Independent study	20

5.4 Graduate attributes

- Critical and creative thinkers who work independently and collaboratively**
 - During assessment, the learner will be required to think critically in order to come up with the creative solution. The assessments that will be given will require critical and effective thinking.
- Knowledgeable practitioners.**
 - The learners must be knowledgeable of the content in order to come up with the required solution.
- Effective communicators.**
 - The learner must be able to express him/herself when answering questions.
- Culturally, environmentally and socially aware within a local and a global context.**
 - The learner must be aware of other ways or methods that can be used to accomplish similar task.
- Active and reflective learners**

During assessment, the learner is expected to demonstrate a deep knowledge of the subject and the assessment will be set in such a way that the student should reflect that expertise.

5.5 Library Orientation

It is assumed that the student did attend the orientation course given by the library staff in their 1st level. This would have highlighted the process of finding books and journals, and other library services and resources. As a product of this the candidate should be in the position to make effective use of the catalogue, use on-line computers and understand referencing styles. All orientation programs will be done at faculty and department level.

6. Scheme of Work

Dates	WEEK	TOPIC
	Week 1	1. Introduction to Platform-based development - Overview of platforms (Web, Mobile, Game, Industrial etc.) - Programming via platform-specific APIs. - Overview of Platform Languages - Programming under platform constraints.
	Week 2	2. Web Platforms - Web programming languages - Web platform constraints - Software as a Service
	Week 3 and 4	3. Mobile Platforms - Mobile Programming Languages - Challenges with mobility and wireless communication - Location-aware applications
	Week 5	- Performance / power tradeoffs - Mobile platform constraints - Emerging Technologies
	Week 6	Recap of Web-Platforms and Mobile-Platforms
	Week 7	
	Week 8 & 9	4. Game Platforms - Types of Game Platforms - Game Platform Languages - Game Platform Constraints
	Week 10	
	Week 11 & 12	5. Industrial Platforms - Types of Industrial Platforms - Robotic Software and its Architecture - Domain Specific Languages - Industrial Platform Constraints
	Week 13	Recap of Game-Platforms and Industrial-Platforms
	Week 14	

7. Copyright and Plagiarism

Presenting someone else's work as your own without proper attribution is a form of intellectual property theft known as plagiarism. Plagiarism, whether intentional or accidental, is strictly prohibited. Accidental plagiarism often occurs through paraphrasing materials from an existing source without proper citation. Therefore, it is essential to exercise caution when paraphrasing and ensure that any borrowed ideas are correctly cited.

It is important to note that most published works are protected by copyright, which legally restricts the copying of material without authorization. Copying copyrighted material without permission is a violation of copyright laws. Should you need to use copyrighted material, it is imperative to obtain the necessary permissions beforehand. Acts of cheating, plagiarism, copying, and unauthorized collaboration are considered serious offenses and will result in disciplinary measures.

Consequences for such actions include, but are not limited to, receiving a zero for the test, assignment, or examination involved, with no opportunity for redoing or submitting the work again. For a comprehensive understanding of the consequences and procedures regarding plagiarism, students are encouraged to consult the plagiarism policy outlined in the DUT general rulebook.

8. Student Support

1. If you require further support relating to the subject content, schedule a consultation with your lecturer within the allocated consultation times. Your lecturer may refer you to one of the support services available.
2. *[Lecturers may mention how they will follow up on students if referred to support services]*
3. You will be able to find a full list of services and comprehensive details for these services on the DUT website http://www.dut.ac.za/student_portal/
4. Below you will find a summarised list of the important support services available to you as students of DUT.

SUPPORT SERVICE	CONTACT DETAILS	WHAT THEY CAN HELP WITH
Faculty Office	East Wing, Hotel School Building, Ritson Campus Mr Lwandile Chiya (Faculty Assistant) Tel: (031) 373 5152 Email: lwandilec@dut.ac.za	Registration, remark/supplementary/special exam application, graduation
Student Counselling & Career Resource Centre Location	Lower Library Complex, Steve Biko Campus, next to financial Aid Reception Desk (031)373 2266 Career Resource Centre (031)373 2571	Individual counselling, career guidance, study skills support
HIV/AIDS Centre	Front Desk (031) 373 2260	Emotional support
Health Clinic	Isolempilo (Gate 5, opposite Sports Centre) 031 373 2223 Ritson Campus (Next to Hotel School Restaurant) 031 3736010	Primary health services, Family Planning, HIV Testing

Financial Aid Services And Scholarships	Lower Library Complex, Steve Biko Campus 031 373 2553 applications@nsfas.org.za NSFAS call centre on 0860 067 327	Application for financial support
Student Residence	Indira Khiramen Tel: 031 373 2494 Email: indirak@dut.ac.za	Application for residence whilst studying at DUT

9. Information and Communications Technology Industry Partnership Committee (ICT-IPC)

The department benefits from the insights and expertise of the ICT-IPC, a specialized board with a critical advisory role. The ICT-IPC is composed of industry experts and professionals who bring a wealth of knowledge and experience in the field of ICT. The primary function of this board is to keep the department informed about the latest trends, innovations, and shifts within the ICT industry. Additionally, the ICT-IPC plays a pivotal role in shaping the educational experience by articulating clear expectations for students as they conclude their academic journey and prepare to enter the workforce. Through its guidance, the ICT-IPC ensures that the curriculum remains relevant and that students are well-equipped with the necessary skills, knowledge, and competencies to thrive in the rapidly evolving ICT landscape. This symbiotic relationship between the ICT-IPC and the department ensures that the academic programs not only meet the current industry standards but also anticipate future developments, thereby preparing graduates for successful careers in the information technology sector.

10. Quality Assurance and Enhancement

Two types of quality assessment evaluation forms are available for use: the Module Evaluation Questionnaire (MEQ) and the Lecturer Evaluation Questionnaire (LEQ). The MEQ focuses on assessing the subject matter itself, while the LEQ is designed to evaluate the teaching and delivery methods of the lecturer. It is crucial for students to provide honest and thoughtful responses on these questionnaires to accurately reflect their experiences with both the subject and the lecturer. Such feedback is invaluable, contributing significantly to the development and improvement of both the course content and teaching methodologies. All feedback received is carefully analysed, and necessary actions are taken to enhance the educational experience.

Students are specifically required to complete the MEQ for the Platform Based Development module and LEQ for the lecturer cited in this study guide.

Throughout the semester, this study guide may be updated with minor revisions as needed to ensure its effectiveness and relevance.